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Stephen M. Westenbroek
U.S. Geological Survey
via Cara Peterman-Phipps, U.S. Geological Survey

Re: Code review of SedSAT3 (Sediment Source Assessment Tool, version 3), IPDS IP-190994

Dear Steve,

Thank you for the careful and generous review of SedSAT3, and for the additional material you sent on October 16. Both were very useful. Your notes on how to read the Kiro-generated documents (as a triage guide, with the line numbers worth checking and the prose taken with a grain of salt) were exactly the right framing, and I have tried to follow it: every concrete finding was checked against the code, real defects were fixed, and findings that turned out not to apply are explained rather than worked around.

All changes are in merge request !4 on code.usgs.gov (branch `review-fixes-oct2026`), organised as one commit per area so they can be reviewed separately:

Commit	Content
`ac1dcd8`	Review notes, test projects, and a regression baseline recorded before any change
`41393e0`	Licensing, attribution and USGS metadata
`ac039a2`	The confirmed code defects
`cdfa4ea`	Seeded, thread-independent random numbers
`68be7ca`	Screening of every imported value
`2f8c463`	Unit tests, regression tests, GitLab CI, optional GA, build cleanup, warnings
`538ed73`	Response document (`docs/review/REVIEW_RESPONSE.md`)
`follow-up`	Follow-up to your October 16 documents: further fixes and tests (Sections 7–11)

In brief:

- All non-USGS license and attribution notices are gone; the repository no longer depends on or points to any personal repository; LICENSE, DISCLAIMER, code.json, CITATION.cff and a third-party license inventory are in place. The license itself, the DOI and the author list await the release coordinator (Section 1).
- Every code defect in your review was confirmed and fixed, plus about a dozen further defects found while doing so (Sections 2 and 6).
- The MCMC now takes a user-set random seed, records it, and gives identical results for the same seed regardless of the number of threads (Section 3).
- Every imported value is screened by a single parser; NaN and Inf are refused at import with the cell location (Section 4).
- There are now 47 unit tests (including the NaN / negative / illegal value tests you suggested), a regression suite on two example projects, and a GitLab CI pipeline running the tests, clang-tidy and cppcheck (Section 5).

Below, each of your comments is reproduced (quoted where short, summarised where long) and followed by my response. Comments are grouped by topic; the source of each is given at the right of its heading: *Email 10/15* and *Email 10/16* are your two emails to Cara, *Summary* is `CODE_REVIEW_SUMMARY_15OCT2026.md`, *Tests* is `TEST_RESULTS.md`, *Observations* is `CODE_REVIEW_OBSERVATIONS.md`, and *Linter* is `linter_output.txt`. File and line references are to the code after the fixes (commit `538ed73`).

Thank you again for your time; I would be very glad to have you take another look once the licensing and ownership questions are settled.

Sincerely,

Arash Massoudieh

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1. Ownership, licensing, attribution and citation

Comment 1.1 *Email 10/15; Summary, threshold question*

In its current form too much of the code and documentation references non-USGS resources. "I'd encourage you to clarify the issue of ownership and commit to either pushing full on for USGS support (i.e. documentation and code and issues migrated to a USGS supported platform) or just continue to develop as is without trying to rebrand as USGS software."

Response. **[Confirm wording with Cara and the release coordinator.]** We intend SedSAT3 to be a USGS software release, as the successor of the USGS Sed_SAT tool (OFR 2017-1062). The repository on code.usgs.gov is now the authoritative copy: code, documentation and issue tracking live there, the README, code.json and CITATION.cff point only to it (placeholder until the final group path is assigned), and no file refers to a personal repository. Following your suggestion, any further development on my side will be submitted as merge requests against the USGS repository.

Status: Repository side done (`41393e0`); institutional decision with the release coordinator.

Comment 1.2 *Email 10/15; Summary, blocker 1*

Several files carry OpenHydroQual / GNU AGPL-3.0 headers with a "purchase a commercial license, contact . . . enviroinformatics.co" clause, contradicting the Utilities submodule's own Apache-2.0 license and far from the USGS CC0 preference.

Response. These files (`TimeSeries.h/.hpp`, `TimeSeriesSet.h/.hpp`) were written by me for another project and reused here. The `Utilities` submodule has been removed and the 23 files SedSAT3 actually builds were brought into the repository (`Utilities/README.md` records where they came from). The AGPL / commercial headers were replaced by the same two-line USGS header that every first-party source file now carries:

```
// SedSAT3 - Sediment Source Assessment Tool, version 3
// U.S. Geological Survey.  See LICENSE.md and DISCLAIMER.md in the repository
root.
```

As the author I release these copies for SedSAT3 under the repository license; I am happy to provide a signed statement to that effect if the release coordinator needs one. Third-party code (QXlsx, QCustomPlot) was not touched.

Status: Fixed (`41393e0`).

Comment 1.3 *Email 10/15; Summary, blocker 2*

The repository is GPL-3.0 because it links GSL, QCustomPlot and Qt Charts, outside the USGS preference (CC0/MIT/Unlicense). Rather than re-engineering, consider the SWB2 approach (a third-party license inventory); the USGS package pfdf was deliberately released under GPL.

Response. I followed your SWB2 suggestion rather than replacing libraries. `THIRD_PARTY_LICENSES.md` lists every dependency (Qt modules, Qt Charts, QCustomPlot, QXlsx, GSL, Armadillo, LAPACK/BLAS, the OpenMP and MSVC runtimes), how SedSAT3 uses each, its license, and where it comes from, and notes that a binary package linking the GPL components is distributed under GPL-compatible terms. `LICENSE.md` currently holds a placeholder with the USGS CC0 text ready. **[License decision: CC0 1.0, MIT, or — following the pfdf precedent — GPL-3.0, which**

would remove the dependency question entirely. To be decided with Cara and the release coordinator; I plan to ask Jonathan King what the pfdf GPL release required.]

Status: Inventory done (`41393e0`); license choice open.

Comment 1.4 *Summary, blocker 3*

The canonical identity, the `Utilities` submodule and the QXlsx fork live under the author's personal GitHub, which points toward non-USGS primary authorship. Raise with the release coordinator.

Response. The `Utilities` submodule is gone (see Comment 1.2) and the QXlsx submodule now points at its public upstream (`QtExcel/QXlsx`), not a personal fork. Authorship is recorded in `CITATION.cff` and the README citation **[author list and order to be confirmed: Massoudieh, A.; Peterman-Phipps, C.; Gellis, A.C.; . . .]**.

Status: Repository side done (`41393e0`, `2f8c463`); authorship for the release coordinator.

Comment 1.5 *Email 10/15*

The README points to GitHub for the code and the user manual, and the suggested citation ("Massoudieh, A., et al. (2023) . . . https://github.com/. . . ") "fails on many levels (a USGS product needs to cite a DOI to a durable, USGS-approved digital repository; all contributors need to be listed)."

Response. The README was rewritten: no GitHub links; the repository is the code.usgs.gov project; the manual points to its USGS location **[manual URL/DOI]**; the citation is in USGS software-release form with the full author list and the release DOI **[10.5066/XXXXXXX]**. The same metadata is in `CITATION.cff` and `code.json`. The README also gained sections on the disclaimer, license, build instructions, build options, tests, and running the portable Windows build.

Status: Fixed with placeholders (`41393e0`).

Comment 1.6 *Summary, should address*

Metadata (`code.json`, README, `createdeb.sh`, submodules) points at personal GitHub; the README's Windows installer link (sedsat.org) serves a different-lineage artifact ("CMBSource" by "EnviroInformatics").

Response. `code.json` was rewritten for code.usgs.gov (and no longer lists the genetic algorithm as a method); `createdeb.sh` now only builds and checksums the Debian package, with no GitHub publishing; the README no longer links to sedsat.org. In the installer project the manufacturer is now "U.S. Geological Survey" and its URLs point to the USGS repository. The Windows package reviewers receive will be the portable build made from this repository (Section 7).

Status: Fixed (`41393e0`).

Comment 1.7 *Summary, optional / polish*

No per-file license headers; no shipped dependency-license (NOTICE) inventory. The README is truncated and reads as adapted from the predecessor "CMBSource" project.

Response. Every first-party source file now carries the USGS header, and `THIRD_PARTY_LICENSES.md` is the dependency inventory. The README was rewritten from scratch (Comment 1.5).

Status: Fixed (`41393e0`).

2. Code defects

Comment 2.1 *Email 10/15; Summary; Tests, Test 7*

Distribution Fitting terminates the application with an unhandled `std::out_of_range` (debugger-traced to `TimeSeriesSet::operator[]` from `ExecuteDistributionFitting`); suspected cause: a non-existent `TimeSeriesSet::append(series, name)` overload, so fit results are stored unnamed.

Response. Confirmed and fixed, though the root cause differs from Kiro's diagnosis: the `append(series, name)` overload exists in this version. The fault was in `ConcentrationSet::CreateFittedDistribution`, which assigned a number to the "Observed" series (`result["Observed"] = maxC/2`). Because `TimeSeries(size_t)` was not `explicit`, the number was silently converted into an empty, unnamed series, and the later lookup of "Observed" threw. Three further faults in the same path were fixed: the log-normal series was silently dropped by a duplicate-name check and `setname(2,...)` then indexed past the end; empty or constant data made the fitting loop infinite (zero step); and the Kolmogorov–Smirnov statistic indexed a series that might not exist. `TimeSeries(size_t)` is now `explicit`, so this class of silent conversion cannot recur; the analysis checks the group, constituent, sample count and positivity and reports a clear message; and `MainWindow::Execute` now catches any exception from any analysis and reports it in a dialog instead of terminating.

Status: Fixed (`ac039a2`). Tests: `distributionFitting_degenerateSets`, `distributionFitting_conductor`; regression (all distribution-fitting steps now succeed).

Comment 2.2 *Email 10/15; Summary; Tests, G1 and G2*

The "Include all elements" checkbox has inverted logic (checking it excludes all tracers), and its label flips between "Include" and "Exclude", which is what produced the inversion. "Just ensure that the behavior of the checkboxes matches your intentions."

Response. Fixed as you suggest: the label is now fixed ("Include all elements"), checked means every element is included, and the box follows the per-element settings (it clears when any element is excluded in the table). `IncludeExcludeAllElements` has no other caller. While writing the test I also found that the element table read the text "No" as *include* (`QVariant("No").toBool()` is true); fixed.

Status: Fixed (`ac039a2`, `2f8c463`). Tests: `includeAll_dataLevel`, `includeAll_checkBox`.

Comment 2.3 *Tests, G3*

Toggling a single element appears to require several clicks.

Response. Agreed that this is friction. The table uses an editor that opens on double-click; making the cell a single-click toggle is a small GUI change that I will make in the next GUI pass.

Status: Open (usability, low).

Comment 2.4 *Email 10/15; Summary; Tests, Test 4c*

Missing `return` on the clean path of `Conductor::CheckNegativeElements` (both overloads) — undefined behaviour, currently masked on MSVC.

Response. This had already been fixed on `main` (`238dd69`) after the reviewed snapshot. I then searched for the same defect everywhere: GCC `-Wreturn-type` found six more functions that could fall off their end, all now fixed. To make this impossible in future, `-Werror=return-type` (GCC/Clang) and `/we4715` (MSVC) are set in every build file. While testing this guard I found a more serious defect in it: `NegativeValueCheck` (and `GrandMean` and `LumpAllProfileSets`) stopped before the last group in alphabetical order, assuming it was the target. With target "Mixture", the source "Pasture" was never checked and the target was checked instead. All three now skip the target by name.

Status: Fixed (`ac039a2`, `2f8c463`). Tests: `negativeCheck_clean`, `negativeCheck_negative`, `negativeCheck_sourceAfterTarget`.

Comment 2.5 *Tests, preprocessing note and Test 4 addendum*

The negative-value guard rejects legitimately negative isotopic tracers (e.g. δ^{13}C) with no transform path, and its message is misleading when the negativity is produced by the organic-matter / size correction rather than present in the input.

Response. The message now distinguishes the two cases: when a value is negative only after the correction, it is reported as "(after the organic matter and size correction; the imported values are positive)". Isotopic tracers such as δ^{13}C are meant to be given the *Isotope* role (with a base element and standard ratio), under which they are handled through their ratio rather than log-transformed; the README now says so in the quick start. Whether an OM/size correction that produces negative concentrations is acceptable is a scientific question I have flagged for the domain review.

Status: Improved (`538ed73`); domain question handed off.

Comment 2.6 *Email 10/15; Summary*

Null dereference in `CBinary::operator[]` (`src/GA/Binary.cpp`).

Response. Fixed: an out-of-range index now throws `std::out_of_range` with the index and size. The same deliberate null dereference was the out-of-range path of `CVector::operator[]` and `CVector_arma::operator[]`, which are used throughout the analyses; those now throw too, so an indexing mistake anywhere produces a message (through the GUI's exception handler) rather than a silent crash.

Status: Fixed (`ac039a2`, `2f8c463`). Test: `binary_indexing`.

Comment 2.7 *Email 10/15; Summary*

Writes to a closed `FILE*` in `src/GA/GA.cpp` (lines 308–309).

Response. `src/GA/GA.cpp` turned out to be a stale copy of the GA that no build compiled (it includes a `StringOP.h` that exists nowhere — the "missing header" you noted); it was removed.

The GA code that is compiled, `include/GA/GA.hpp`, now opens its files through a `unique_ptr` handle that closes exactly once and skips writing if `fopen` fails; a `printf` with three `%le` and one argument and an unsynchronised counter in its OpenMP loop were also fixed. Every other `fopen` in the code now checks for failure.

Status: Fixed (`ac039a2`, `538ed73`).

Comment 2.8 *Email 10/15; Summary*

Possible `log()` of a non-positive value in the Box–Cox transform (`ApplyBoxCoxTransform`).

Response. Confirmed for zero values (negative values were already skipped; zeros gave $-\infty$). Any constituent with a zero or negative value is now left untransformed, and the analysis lists the constituents it skipped. $\lambda = 0$ is handled as the log case; for $|\lambda| < 0.01$ the transform is computed as $\mathrm{expm1}(\lambda \log x)/\lambda$ so it keeps full precision; elsewhere the formula is unchanged, so optimal λ values do not move. Two related faults were fixed: outlier detection used its own copy of the formula, which scored skipped constituents from $x - 1$ and so shifted their z-scores by $1/\sigma$; and the (unused) Box–Cox log-likelihood took the log of a sum that was always negative.

Status: Fixed (`ac039a2`). Tests: `boxcox_positiveData`, `boxcox_zeroAndNegativeData`, `boxcox_lambdaZeroAndNearZero`, `boxcox_logLikelihoodIsFinite`.

Comment 2.9 *Email 10/15; Summary*

Widespread uninitialised variables flagged by clang-tidy and cppcheck.

Response. All 128 findings of clang-tidy's `cppcoreguidelines-init-variables` and `cppcoreguidelines-pro-type-member-init` were fixed: in-class initialisers for members, and NaN for floating-point locals so that any read-before-write is visible rather than silent. One was a real defect: the stepwise DFA could report an unassigned Wilks' λ when no element improved the fit (it now reports NaN). These two checks are errors in the CI clang-tidy job, so new instances fail the pipeline.

Status: Fixed (`ac039a2`); enforced in CI.

3. Reproducibility and random numbers

Comment 3.1 *Email 10/15, item 1; Summary; Tests, Test 1*

The MCMC is non-reproducible: identical inputs give materially different outputs (Bank upper credible bound 0.47–0.67 over three runs), there is no seed control and no recorded seed. "Consider adding a way to provide a user-defined seed to allow for testing."

Response. Done. CMB Bayesian, CMB Bayesian-Batch and Test CMB Bayesian have a new *Random seed* field (blank = drawn from the operating system). The seed used is shown in the name of the results ("MCMC results for 'X' (random seed N)"), stored with the project and in the command-line report, and written on the first line of the samples file. The same seed and settings give byte-identical results. The generator is `std::mt19937_64` seeded through `std::seed_seq`; the uniform and normal values are computed from its raw output in `include/randomstream.h` rather

than with `std::*_distribution`, whose algorithms differ between standard libraries, so a recorded seed reproduces a run on Linux, macOS and Windows.

Status: Fixed (`cdfa4ea`). Tests: `mcmc_sameSeedSameResult`, `mcmc_differentSeedDifferentResult`, `random_streamsAreReproducible`, `random_seedParsing`.

Comment 3.2 *Email 10/15, item 2; Summary; RABBIT_HOLE_rand_and_threads.md*

`rand()`/`srand()` called inside an OpenMP parallel region is a data race that also degrades the generator's statistical properties.

Response. Kiro was right that this was a race. Each chain now has its own stream, `RandomStream(seed, chain)`, used for its random start and every step, so no generator is shared between threads and the result does not depend on scheduling: the samples are identical with 1, 3 or 8 threads. I audited every OpenMP region: the MCMC step loop also wrote a shared cached maximum from every thread (now a named critical section), and the GA loop's counter is now atomic. Every other `rand()`, `srand()` and clock-seeded generator in the code now draws from a per-thread stream with a fixed seed, and Error Analysis (which removes random samples) now has its own seed field like the MCMC. clang-tidy's `concurrency-mt-unsafe` check is now an error in CI, so `rand()` cannot creep back in.

On the details of `RABBIT_HOLE_rand_and_threads.md` (your instinct not to bet the house was sound): the conclusion and the remedy are right, but the mechanism is overstated for the platforms SedSAT3 runs on. The `srand` block only ran on Windows (`#ifdef WIN64`), where the MSVC runtime keeps `rand()` state per thread, and glibc's `rand()` takes a lock, so "lost updates" were theoretical. The real problems were that the order of draws depended on thread scheduling, Windows reseeded from the clock every iteration, MSVC's `RAND_MAX` of 32767 gave uniforms only 15 bits of resolution, and Linux and Windows produced different sequences. Its suggested fix also needed two corrections, both made: `std::normal_distribution` is not portable across standard libraries, and streams must be per *chain*, not per thread, for results to be independent of the thread count.

Status: Fixed (`cdfa4ea`). Test: `mcmc_independentOfThreadCount`.

Comment 3.3 *Tests, Test 1 (convergence) and your note on it*

Increasing samples from 500 to 3000 reduced, but did not remove, the run-to-run spread, and the software surfaces no convergence diagnostics (R-hat, effective sample size, trace plots). [Your note: "that conclusion is a little strong."]

Response. Agreed on both counts. Part of the spread came from a units issue in the dialog: *Number of samples* and the burn-in count samples across *all* chains, so 3000 samples with 8 chains is only 375 steps per chain; at 40 000 samples with 20 000 burn-in the validation targets in our test project are recovered at 85–99% to the correct source, in agreement with Levenberg–Marquardt. Now that runs are reproducible, sampling noise and non-convergence can at least be separated by re-running with the same seed and more samples. Exposing R-hat, effective sample size and trace plots is a feature I plan for a later version; whether the current defaults are adequate is for the domain review.

Status: Partly addressed (reproducibility); diagnostics deferred; domain question handed off.

Comment 3.4 *Tests, LM contrast*

The deterministic Levenberg–Marquardt method reproduces exactly, which shows the variation is specific to the stochastic path.

Response. On checking, LM was less deterministic than it looked: its random starting point came from unseeded `rand()`, so it repeated only while nothing reseeded the generator (the MCMC did, on Windows) and differed between Linux and Windows. It now draws its starting point from a fixed-seed stream created for each solve, so it is repeatable everywhere. Most results changed by less than 10^{-5}, but for one target sample in the domain-review project they changed by up to 0.06, showing that the LM solution for that sample depends on its starting point. I have flagged this for the domain review.

Status: Fixed (`cdfa4ea`); regression baseline updated with explanation.

4. Input validation

Comment 4.1 *Email 10/15, item 3; Tests, Test 4b; Email 10/16*

A bare "NaN" passes `QString::toDouble` and is stored raw; whether it then affects a result depends on whether each consumer calls `removeNaNs()` or checks `isfinite`, which is inconsistent. "Have Claude help create some unit tests that would go after the odd NaN / legitimately negative / otherwise illegal values."

Response. There is now one parser, `include/numericparsing.h` (`parseNumericCell` and overloads for spreadsheet and project-file values), used by the Excel import, project loading, the element table and the command line. It refuses empty cells, NaN / Inf / Infinity in any case, overflow, text, and decimal commas or thousands separators, each with a specific reason. A refused value stops the import with sheet, row, column and value (Excel) or group, sample and constituent (project file) — the same treatment empty and non-numeric cells already had, since SedSAT3 has no missing-data value. Two findings from tracing the behaviour you observed: the "Ignored NaN, Inf, or -Inf value" message came from Qt Charts while plotting, so the NaN *had* been stored; and project loading turned a missing value or a NaN (which Qt saves as `null`) into 0, and silently dropped a sample that failed to load. Both now stop the load with the location. The only ad-hoc data guard downstream (the Box–Cox optimiser's `min == min` test) uses a shared helper; the other `isfinite` uses guard computed quantities, not data, and `removeNaNs()` has no callers. The unit tests you suggested cover 8 valid formats and 14 illegal inputs, spreadsheet and JSON values, a saved-and-reloaded NaN, zero and negative data in Box–Cox, and negative values in the source groups.

Status: Fixed (`68be7ca`). Tests: `parser_acceptsNumbers`, `parser_rejects`, `parser_variantAndJson`, `boxcox_zeroAndNegativeData`, `negativeCheck_*`.

Comment 4.2 *Tests, Test 5*

Project persistence round-trips correctly (data, tracer exclusions and MCMC results survive save and reload).

Response. Thank you for checking. Saved projects remain backward compatible: no serialised key was renamed, and projects with only finite values load exactly as before.

Status: No change needed.

5. Tests, CI, the genetic algorithm, and hygiene

Comment 5.1 *Email 10/15, item 4; Summary*

There appear to be no automated tests or CI beyond a docs workflow; add at least one end-to-end regression against a known result and run the static-analysis tasks in CI.

Response. Added. `tests/unit/` has 47 Qt Test checks covering every defect above. `tests/regression/` runs the deterministic analyses (Box–Cox, outliers, Kolmogorov–Smirnov, distribution fitting, ANOVA, correlation and covariance, bracketing, DFA, EDP, Levenberg–Marquardt batch) on two example projects and compares them with a stored baseline at a relative tolerance of 10^{-9}; the output of the code before the fixes is kept alongside, so every change the fixes made is traceable. `.gitlab-ci.yml` builds with CMake (GUI, command line, tests), with the GA enabled, and with the qmake projects; runs the unit and regression tests; runs clang-tidy (initialisation checks as errors) and cppcheck; has a manual Windows job for a Windows runner; and publishes the Doxygen documentation to GitLab Pages. The existing GitHub documentation workflow is kept for the development mirror.

Status: Fixed (`2f8c463`).

Comment 5.2 *Email 10/15, item 5; Summary; Tests, Test 6*

The genetic algorithm is advertised in the README and code.json but is in no menu and does not compile (missing `StringOP.h`); either wire it up or drop the claim. `MCMC.h` includes `GA.h`, so the GA header is not cleanly deletable.

Response. The GA works and is deliberately not offered: Levenberg–Marquardt solves the same problem more efficiently, and two deterministic methods would confuse users. It is now kept in the source but excluded from the default build (`CONFIG+=ga` with qmake, `-DENABLE_GA=ON` with CMake); both settings build and CI builds both. `MCMC.h` no longer includes `GA.h`. The README and code.json no longer list it as a method, the README says it is included but disabled and why, and the command-line program and its manual leave out its commands. The missing `StringOP.h` was only included by the stale `GA.cpp` (Comment 2.7).

Status: Fixed (`2f8c463`). CI job `build:linux-ga`.

Comment 5.3 *Email 10/15, item 6; Summary*

A dead submodule URL blocks fresh clones; stray build artifacts; spelling in identifiers (`purturbation`, `FileBrowserPushButtom`, “samle”).

Response. The QXlsx submodule URL works (it redirects to the upstream `QtExcel/QXlsx`, and the pinned commit is fetchable); it now points at the upstream directly. The failing clone was most likely the `Utilities` submodule, which is gone. A fresh `git clone -recursive` of the branch builds. Removed: `wasm_bin/` build output, the stale `CMBSource.pro` (which listed files that do not exist), a stray `gitignore` copy and an already-applied script; the Linux build no longer links a prebuilt `libQXlsx.a` but compiles QXlsx from the submodule; `.gitignore` covers build folders, IDE files and binaries. Spelling: `FileBrowserPushButton`, `perturbation` (the old MCMC setting

names are still accepted), “sample”, “Description” (the misspelled form appeared as a label in the GUI) and `sizeconstituent`.

Status: Fixed (`2f8c463`).

Comment 5.4 *Summary, optional / polish*

Three coexisting build systems with no documented source of truth; repo layout mid-transition (non-GUI core at the repository root).

Response. The qmake projects now share one file, `sedsat3_core.pri`, for the analysis sources, libraries and options; CMake builds the GUI, the command line program and the tests from the same source list; the Visual Studio project is kept for Windows developers; the README documents all three. I have left the move of the root sources under `src/` for a separate change, since it is large and mechanical and would make this review harder to follow.

Status: Improved (`2f8c463`); layout change deferred.

Comment 5.5 *Email 10/15; Email 10/16; Linter*

Compile with multiple compilers with warnings “turned up to 11” and systematically reduce them; the more significant warnings generally point at minor defects.

Response. I did this with GCC 13 (and clang-tidy from LLVM 20). Unique first-party warnings in the GUI build went from 244 to 101 with `-Wall -Wextra`, and from 618 to 449 with `-Wall -Wextra -Wpedantic -Wshadow -Wconversion`. As you predicted, the significant ones were real defects: two operators that called themselves (stack overflow), a constructor that wrote past the end of an empty matrix, a function returning a reference to a local, printf format mismatches, invalid escapes, and the missing returns. Every `-Wreturn-type`, `-Winfinite-recursion`, `-Wreturn-local-addr`, `-Wformat`, `-Wmaybe-uninitialized`, `-Wmisleading-indentation`, `-Wreorder`, unused-variable/parameter and copy-constructor warning is gone. The remainder are implicit integer/floating conversions (`-Wconversion` 293, `-Wsign-compare` 100, `-Wfloat-conversion` 20) and `-Wshadow` 35 (mostly constructor parameters named after members); I prefer to change the numerical conversions one at a time with the regression tests rather than in bulk. The clang-tidy findings in `linter_output.txt` are answered in Section 9.

Status: Largely fixed (`2f8c463`).

6. Further defects found and fixed during the response

Comment 6.1 *Found while addressing the review*

Defects not in the review, found while fixing the items above.

Response.

- ANOVA read the “Use only selected samples/elements” options and then ignored them (it always used the selected samples and every element).
- `NegativeValueCheck`, `GrandMean` and `LumpAllProfileSets` skipped the source sorting after the target (Comment 2.4); ANOVA’s total sum of squares left that source out.

- `CMBTimeSeries::operator=` and `CVector_arma::operator*=` called themselves (stack overflow on use).
- `CMatrix_arma(const CVector_arma&)` built a temporary instead of sizing the matrix, then wrote past its end.
- `CMatrix_arma_sp::get` returned a reference to a local, so writes to sparse matrices were lost.
- `RangeSet`'s copy constructor copied nothing; the copy constructors of several result types dropped their notes and axis labels.
- The MCMC's shared prediction rows were written from every thread (Comment 3.2).
- `TimeSeries::addPoint` did not compare the first two intervals, so an uneven series could be treated as evenly spaced.
- `gamma()` clashed with `::gamma` under Clang.

Status: Fixed (`ac039a2`, `2f8c463`).

Comment 6.2 *Effect on results*

How do the fixes change the results the software produces?

Response. The regression suite was run on the code before the fixes and after; every difference is explained in `docs/review/REVIEW_RESPONSE.md`. In short: distribution fitting now succeeds where it crashed; outlier z-scores of constituents with negative values (C13; N15 in Forest) shift by a constant $1/\sigma$; Box–Cox and outlier analyses list the constituents they skip; Levenberg–Marquardt moves by less than 10^{-5} for most samples and up to 0.06 for one (Comment 3.4); and ANOVA p-values change, some across 0.05, because the selection options now take effect and the previously dropped source group is included. Box–Cox λ, Kolmogorov–Smirnov, correlation and covariance, bracketing, DFA and EDP are unchanged. The ANOVA and Levenberg–Marquardt changes have been passed to the domain reviewers.

7. Windows build and the portable package

Comment 7.1 *Email 10/16; BUILDING_ON_WINDOWS_WITH_PIXI.md*

The reviewed copy had no viable Windows compilation path; Kiro created one with pixi (conda-forge libraries driven from `pixi.toml`, with `pixi run lint` for the strict-warning build). "This may or may not be useful."

Response. It is useful, and since it is how you build, I want it to work out of the box at the re-review. Several of the pitfalls your guide works around are now gone from the repository: the QXlsx URL is fixed, `Utilities` is no longer an (empty) submodule, the hard-coded `#define ARMA_USE_SUPERLU` that broke the win-64 link has been removed (SuperLU is now used only if the installed Armadillo provides it), MSVC gets `/utf-8` from CMake (your C4828), and the duplicate `USE_QCHARTS` definition is guarded (your C4005). The guide itself needs three adjustments to match the repository: its `lint` task lists the removed `src/GA/GA.cpp`; its `.clang-tidy` would replace the repository's stricter one (which CI uses, with the initialisation and `rand()` checks as errors); and `deploy-win` uses `-compiler-runtime` and LLVM's OpenMP DLLs, whereas MSVC builds need `vcomp140.dll` (see `docs/PACKAGE-Windows-Portable.md`). **[Plan: commit a corrected pixi.toml (configure/build/test/lint/cppcheck/deploy tasks driving the existing CMake build) after testing it on Windows, so that 'pixi run build test lint' works on your machine; this will also give the first MSVC /W4 and cppcheck results.]** The Visual

Studio project now compiles the vendored QCustomPlot instead of a copy outside the repository; it still links a prebuilt QXlsx library, which I will change to the submodule as the qmake and CMake builds already do. The portable, no-administrator Windows package (unzip and run) will be built from the merged code and sent to the reviewers.

Status: Build fixes done; `pixi.toml` pending a Windows test.

8. Detailed observations (CODE_REVIEW_OBSERVATIONS.md)

Comment 8.1 *Observations, §1–16*

The full review log, "nuggets of gold mixed with Kiro fluff".

Response. I checked every concrete finding against the code. Most were already addressed by the changes above; the rest are listed here with what was done. A few did not hold up when checked, and I note them so the record is accurate, not to diminish the review: they were all worth looking at.

§	Finding	Response
1	No command-line entry point	A command-line program, `sedsat3-cli`, runs any analysis from a JSON script; the regression tests use it.
1, 9	GA listed as a method; GA not built	Fixed (Comment 5.2).
2	Three build systems, no source of truth	Improved (Comment 5.4).
2	`CMBSource.pro` hard-codes a personal path	File removed.
2	The same kind of paths in `SedSat3.vcxproj`	QCustomPlot now from `thirdparty/`; the prebuilt QXlsx still to change (Section 7).
2	Hygiene script not run; stray `gitignore`	Both removed.
2	`CMB_Source.aip` still tracked	Kept on purpose (the installer) and rebranded to the USGS.
3a	QXlsx submodule URL dead	Fixed (Comment 5.3).
3b	`Utilities` is a personal-namespace submodule	Brought into the repository (Comment 1.2).
3b	Hard-coded `ARMA_USE_SUPERLU` breaks Windows	Fixed (Section 7).
3b	No Windows build path in the repository	MinGW guide, Visual Studio project, CMake, and a manual Windows CI job; pixi to follow (Section 7).
3c	qcustomplot6 version not documented	`THIRD_PARTY_LICENSES.md` (2.1.1).
4, 4a	README truncated; advertises sedsat.org, GA, GitHub	Rewritten (Comments 1.5, 1.6).
4	Dependency versions unpinned	README and `THIRD_PARTY_LICENSES.md` give versions (Qt 6.2+, 6.8 tested, . . .).
4a	"CMB"-era artifacts	Removed (only icon file names remain).
5	No automated tests or CI	Fixed (Comment 5.1).
5, 9	`testmcmc` is a non-functional stub	It is the model behind "Test CMB Bayesian", and is now the fixture of the MCMC reproducibility tests.
6	MSVC C4828 (non-UTF-8 bytes)	Four files were Latin-1; all sources are now UTF-8, and CMake passes `/utf-8` to MSVC.

§	Finding	Response
6	MSVC C4005, C4834	C4005: the `USE_QCHARTS` and `NOMINMAX` redefinitions are now guarded. C4834 not reproduced (no MSVC here); will check with the first MSVC `/W4` run.
9	Non-GUI core at the root; mixed header layout	Deferred to a separate change (Comment 5.4).
9	`FileBrowserPushButtom` misspelling	Fixed.
9	Duplicate `FilePushButton` class	Unused; removed.
9	Overlapping chart layers (Qt Charts and QCustomPlot)	Acknowledged; to be resolved together with the GPL-dependency decision (Comment 1.3).
9	Orphan `drand.cpp`; empty `GA_impl` files	Removed.
10	Clock-seeded `srand`; no seed control or record	Fixed (Comment 3.1).
10	Every thread's default stream starts from the same seed	Documented: parallel code must use its own seeded stream, as the MCMC does; no parallel code uses the default stream.
11.1–11.2	Platform-specific seeding; `rand()` race in OpenMP	Fixed (Comment 3.2).
11.3	GA file I/O inside an OpenMP critical section	Correct but slow; left as is because the GA is off by default.
11.4	`qDebug` in the MCMC loop	Removed.
11	OpenMP use not documented	The chains run in parallel and results do not depend on the thread count; noted in `include/randomstream.h` and the README.
12	Project declares clang-tidy/cppcheck in `pixi.toml`	There was no `pixi.toml` in the repository; this was your local environment. Both tools now run in CI.
12a.1	Null dereference in `CBinary::operator[]`	Fixed (Comment 2.6).
12a.2	Writes to a closed `FILE*`	Fixed (Comment 2.7).
12a.3	Missing return in both `CheckNegativeElements` overloads	Fixed; in the reviewed snapshot only the map overload lacked a return, and its result was not used, so the guard itself worked (Comment 2.4).
12a.4	`log()` of a non-positive value in `ApplyBoxCoxTransform`	Fixed; the always-negative `log` argument was in `CalculateBoxCoxLogLikelihood` (Comment 2.8).
12b	`Interface` limits uninitialised; copy operations incomplete	Fixed: members initialised, and the copy constructor and assignment now copy every member, so copies keep their notes (outlier and bracketing flags) and highlight limits.
12b	Uninitialised members of `Observation`, `Parameter`, `element_information`	Fixed (Comment 2.9).
12b, 12d	Performance and implicit-widening notes	See Section 9.
12d	Missing `StringOP.h`; GA headers use incomplete types	Fixed; GA.hpp now includes what it uses, and the GA builds in CI.
12d	`MCMC.h` includes `GA.h`	Fixed.
13	GPL-3.0 outbound license; GPL dependencies	Comment 1.3.
13	GSL used for a single `gsl_multifit_wlinear` call	GSL is used more widely (CDFs, statistics, matrices in about ten files), so replacing it is not a one-call change; another reason to prefer the inventory approach.

§	Finding	Response
13, 13a	No per-file headers; AGPL headers in `Utilities`	Fixed (Comments 1.2, 1.7).
14, 14a	Metadata and installer point outside the USGS	Fixed (Comment 1.6); the parallel sedsat.org distribution is for the release coordinator.
14	`createdmg.sh` version stale	Updated to 1.1.7.
16	Spelling; add codespell to CI	Spelling fixed (Comment 5.3); codespell will be added to CI.
8.4	Distribution Fitting crash caused by a missing overload	Fixed; the cause was an implicit conversion (Comment 2.1).
8.6	"Include all elements" inverted	Fixed (Comment 2.2).

Comment 8.2 *Found while checking the observations*

Further defects found while verifying the findings above.

Response.

- `ObservedDataforSelectedSample_Isotope` read the base element from an empty sample name in one branch, dereferencing a null pointer.
- The two-sample t-test computed its degrees of freedom as an unsigned value that wrapped around with fewer than two samples in a group; it now reports NaN.
- The distribution curve for uniform and Dirichlet priors consisted of NaN points, and a one-point curve divided by zero.
- The results table compared the *time* rather than the value with the upper highlight limit.
- An MCMC initialisation routine computed a step from $-\text{low} - \text{high}$ instead of $\text{high} - \text{low}$.
- `CGA::operator=` took a non-const reference and returned by value.

All are fixed. Unit tests were added for the uniform curve and for copies keeping their notes.

9. clang-tidy findings (linter_output.txt)

Comment 9.1 *Linter*

`pixi run lint` output: clang-tidy over 13 translation units with `bugprone-*`, `clang-analyzer-*`, `concurrency-*`, `cppcoreguidelines-init-variables`, `-pro-type-member-init`, `performance-*` and `modernize-*` checks (101 first-party diagnostics, 8 of them errors).

Response. Every error and every finding that indicated a defect is fixed or no longer applies:

Check	Count	Status
`cppcoreguidelines-init-variables`	35	Fixed (Comment 2.9); one in the old `GA.cpp` was a false positive.
`cppcoreguidelines-pro-type-member-init`	13	Fixed with in-class initialisers.
`clang-diagnostic-error`: `StringOP.h` not found	1	File removed (Comment 2.7).

Check	Count	Status
`clang-diagnostic-error`: incomplete types in `GA.hpp`	7	Fixed: `GA.hpp` includes `progressreporter.h` itself, and its `Q_version` guards (a macro defined nowhere) are now `Q_GUI_SUPPORT`.
`clang-analyzer-cplusplus.NewDeleteLeaks` (progress windows)	5	Fixed: progress windows are owned by a scoped object.
`concurrency-mt-unsafe` (`rand`)	2	Fixed (Comment 3.2); now an error in CI.
`clang-analyzer-deadcode.DeadStores` (ANOVA options)	2	Fixed: the options were ignored (Section 6).
`UndefinedBinaryOperator`, `CallAndMessage`	4	Fixed (initialisation, uniform curve).
`UndefReturn`, `modernize-use-nullptr` (`CBinary`)	2	Fixed (Comment 2.6).
`bugprone-unhandled-self-assignment`	1	Fixed (`CGA::operator=`).
`bugprone-branch-clone` (GA parameter mapping)	1	**[Identical branches; confirm intent and simplify]** (GA only).
`bugprone-integer-division`	1	Intended (the result is bounded below by 1).
`bugprone-float-loop-counter`, `FloatLoopCounter`	6	Grid loops over x and λ; a zero step is now impossible (Comment 2.1). Kept.
`bugprone-throwing-static-initialization`	1	False positive (`atan(1.0)`).
`bugprone-implicit-widening-of-multiplication-result`	4	Counts are small integers; to be cast explicitly.
`performance-unnecessary-value-param`, `performance-for-range-copy`	15	Style; to be cleared, then the `performance-*` checks enabled in CI.

10. Reviewer GUI test plan (REVIEWER_TEST_PLAN.md)

Comment 10.1 *REVIEWER_TEST_PLAN.md and TEST_RESULTS.md*

Tests 1–7 and the GUI observations G1–G3.

Test	Status and automated test
1. MCMC reproducibility	Fixed; `mcmc_sameSeedSameResult`, `mcmc_differentSeedDifferentResult`. Convergence diagnostics deferred (Comment 3.3).
1b. LM determinism	Made genuinely repeatable (Comment 3.4); regression suite.
2. Sum to one, non-negative	Now tested for both parameterisations: `levenbergMarquardt_contributionsSumToOne`. The Bayesian contributions satisfy it by construction (the last is one minus the others).
3. LM vs Bayesian	Not run by the review; left to the domain review.
4a. Required column removed	Import refuses a missing cell with its location (existing behaviour); **[run and record]**.
4b. NaN / non-numeric cell	Fixed (Comment 4.1); parser tests.
4c. Negative values	Fixed (Comment 2.4); `negativeCheck_*`.
4c addendum	Message improved (Comment 2.5).
4d. Single-sample source	Now tested: `levenbergMarquardt_singleSampleSource`, `distributionFitting_conductor`.
5. Round trip	Passed in your test; unchanged.
6. GA advertised	Fixed (Comment 5.2).
7. RNG thread safety	Fixed; `mcmc_independentOfThreadCount`.

Test	Status and automated test
Test 7 (results). Distribution Fitting crash	Fixed (Comment 2.1).
G1–G3	G1, G2 fixed; G3 open (Comments 2.2, 2.3).

The test plan's notes that "SedSAT3 has no CLI" and that it is launched with `pixi run run-app` are now out of date; Tests 1 and 7 can be scripted with `sedsat3-cli`.

11. USGS software release checklist

Comment 11.1 *software-release-checklist.pdf*

The USGS checklist for open-source and scientific-information-product releases.

Response. The repository items are in place; the remaining items are decisions or process steps for the release coordinator, Cara and me:

Item	Status	Remaining
README.md, DISCLAIMER.md (provisional)	Done	Placeholders: repository URL, manual, citation details.
LICENSE.md	Placeholder	License choice (Comment 1.3).
code.json	Done, placeholders	Version, contact, DOI, URLs; `name` to match the GitLab project; `downloadURL` to point at `main`; **[ask whether the AI tag applies to AI-assisted development]**.
Project in a USGS group (not a personal namespace)	Not done	Release coordinator: move from `cphipps/` .
Default branch is what is released	In progress	Merge request !4 into `main`.
Administrative security review of every commit	Process	Security reviewer; the history includes old binaries and the AGPL-era files; decide whether to publish or rewrite it.
IPDS record; reviews logged	Process	Cara: log this review, this response and the re-review.
DOI reserved	Process	Release coordinator.
Release-candidate branch named for the version	Not done	After the version is chosen (semantic version, no "v").
Domain review	Open	LM start sensitivity, MCMC convergence, OM/size correction producing negatives, isotopic tracers, ANOVA changes.
Final changes on the release branch (approved disclaimer, tagged URLs)	Not done	At release.